

Bhairahawa Multiple Campus

Tribhuvan University

Bhairahawa ,Rupandehi ,Nepal



Database Management System

BIT 3rd Semester (2082)

Submitted by

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Submitted To

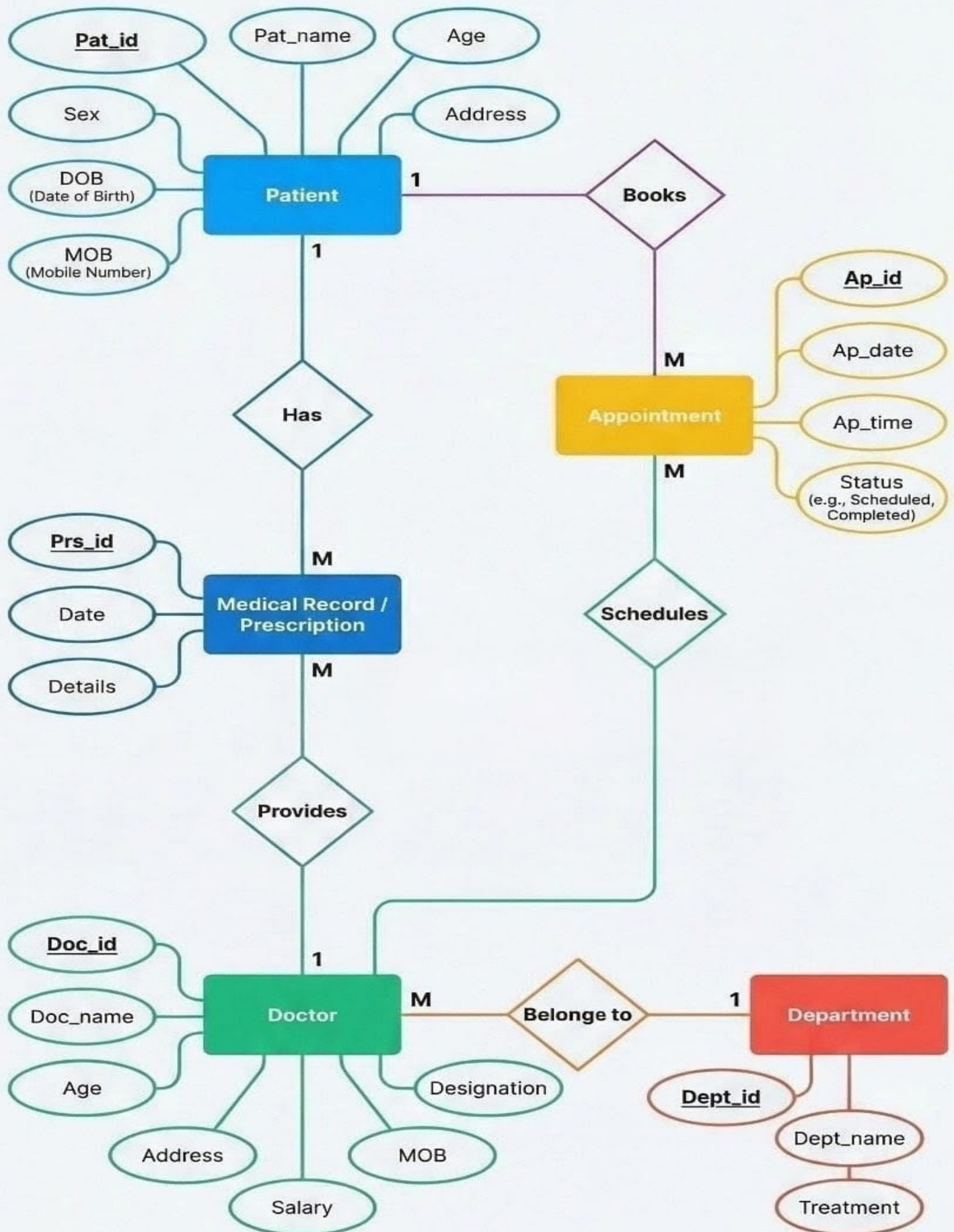
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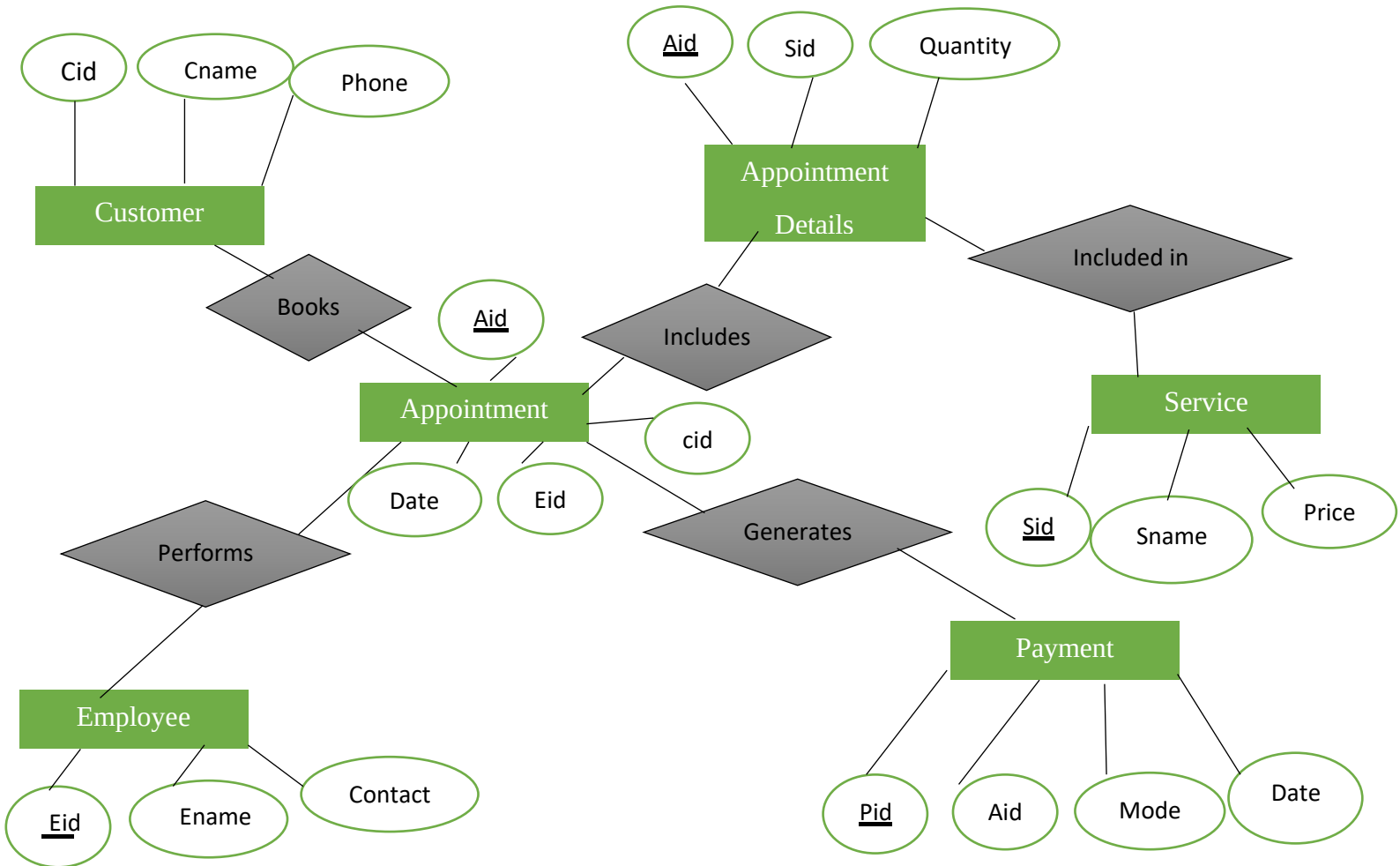
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1. Design the ER diagram for hospital and assumptions at least five entities with attributes.



Q2. Design the ER Diagram for Nationwide Electronic Shop and assumptions at least five entities with attributes



LAB 1: DDL

Q3. Describe about the database and DBMS with example.

An electronically stored, systematic collection of data that can include words, numbers, images, videos, and other types of files is known as database. It provides us with organized data storage, efficient access, security and scalability. The components of database include data, schema, queries, users etc. The types of databases include relational database, object-oriented database, hierarchical database, network database etc. Examples: a library database, a bank's database, an online shopping platform, etc.

The software that interacts with databases to manage data systematically is called Database Management System. It acts as an interface between the user, applications, and the database. Some features of DBMS include collection of interrelated data, interface for operations, custom database creation, query processing, etc. DBMS has various advantages and is very essential for data management and organization. However, some of the challenges faced by DBMS are:

- a) High cost
- b) Complexity
- c) Performance overhead
- d) Frequent Updates and maintenance
- e) Security risks
- f) Risk of data corruption or loss

Examples: MySQL, Oracle, Cassandra, MongoDB, etc.

Q.N.4. According to all Entities which are considered in ER diagram, Create all tables with primary key and foreign key nationwidedb database.

Query :

Create database

nationwidedb;

Use nationwidedb;

Create table customer (

customerid int primary key

name varchar(100),

```
    phone varchar(15),  
    email varchar(100)  
);
```

```
Create table employee (  
    employeeid int primary key,  
    name varchar(100),  
    role varchar(50),  
    contact varchar(100)  
);
```

```
Create table service (  
    serviceid int primary key,  
    servicename varchar(100),  
    price decimal(10, 2),  
    duration int -- duration in minutes);
```

```
Create table appointment (  
    appointmentid int primary key,  
    customerid int,  
    employeeid int,  
    date date,  
    time time,  
    foreign key (customerid) references customer(customerid),  
    foreign key (employeeid) references employee(employeeid)  
);
```

Create table appointmentdetails (

appointmentid int,

serviceid int,

quantity int,

primary key (appointmentid, serviceid),

foreign key (appointmentid) references appointment(appointmentid),

foreign key (serviceid) references service(serviceid)

);

Create table payment (

paymentid int primary key,

appointmentid int,

mode varchar(50),

amount decimal(10, 2),

paymentdate date,

foreign key (appointmentid) references appointment(appointmentid)

);

Result:

```
MariaDB [nationwidedb]> show tables;  
+-----+  
| Tables_in_nationwidedb |  
+-----+  
| appointment  
| appointmentdetails  
| customer  
| employee  
| payment  
| service  
+-----+
```

Q.N.5. Insert minimum 5 records in each table to check functionality of foreign key & primary key and join operations.

i. Primary key:

Query: show create table customer;

Result:

```
customer | CREATE TABLE `customer` (
  `CustomerID` int(11) NOT NULL,
  `Name` varchar(100) DEFAULT NULL,
  `Phone` varchar(15) DEFAULT NULL,
  `Email` varchar(100) DEFAULT NULL,
  PRIMARY KEY (`CustomerID`)
```

ii. Foreign Key:

Query: show create table payment;

Result:

```
payment | CREATE TABLE `payment` (
  `PaymentID` int(11) NOT NULL,
  `AppointmentID` int(11) DEFAULT NULL,
  `Mode` varchar(50) DEFAULT NULL,
  `Amount` decimal(10,2) DEFAULT NULL,
  `PaymentDate` date DEFAULT NULL,
  PRIMARY KEY (`PaymentID`),
  KEY `AppointmentID` (`AppointmentID`),
  CONSTRAINT `payment_ibfk_1` FOREIGN KEY (`AppointmentID`) REFERENCES `appointment` (`AppointmentID`)
```

iii. Join Operations:

select c.name, a.appointmentid, s.servicename, p.amount

from customer c

join appointment a on c.customerid = a.customerid

join appointmentdetails ad on a.appointmentid = ad.appointmentid

join service s on ad.serviceid = s.serviceid

join payment p on a.appointmentid = p.appointmentid;

Result:

Name	AppointmentID	ServiceName	Amount
Ram Shrestha	301	Laptop Repair	7000.00
Ram Shrestha	301	Software Installation	7000.00
Sita Thapa	302	Phone Screen Replacement	6000.00
Hari Gurung	303	Battery Replacement	2500.00
Gita Magar	304	Data Recovery	7000.00

Q.N.6.Create table employee with minimum six attributes.

Query:

```
create table employee001 (
    employeeid int primary key,
    name varchar(100),
    role varchar(50),
    contact varchar(100),
    address varchar(200),
    salary decimal(10, 2)
);
```

Result:

```
Employee001 | CREATE TABLE `employee001` (
  `EmployeeID` int(11) NOT NULL,
  `Name` varchar(100) DEFAULT NULL,
  `Role` varchar(50) DEFAULT NULL,
  `Contact` varchar(100) DEFAULT NULL,
  `Address` varchar(200) DEFAULT NULL,
  `Salary` decimal(10,2) DEFAULT NULL,
  PRIMARY KEY (`EmployeeID`))
ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci
```

Q.N.7. Rename table employee with new name employees.

Query:

```
alter table employee001 rename to employees;
```

Result:

Tables_in_nationwidedb
appointment
appointmentdetails
customer
employee
employees

Q.N.8. Write SQL statement for listing all tables.

Query: show tables;

Result:

Tables_in_nationwidedb
appointment
appointmentdetails
customer
employee
employees
payment
service

Q.N.9. Write a query to delete all employee record who are from

Humla . Query: delete from employees where address = 'humla';

Result:

EmployeeID	Name	Role	Contact	Address	Salary
108	Mina KC	Manager	mina@shop.com	Kathmandu	45000.00
109	Hari Bhandari	Receptionist	hari@shop.com	Pokhara	30000.00
110	Gita Shrestha	Technician	gita@shop.com	Biratnagar	38000.00

Q.N.10. Write a query to update employee's name Geeta with Sita.

Query: update employees set name = 'sita shrestha' where name = 'gita shrestha';

Result:

EmployeeID	Name	Role	Contact	Address	Salary
108	Mina KC	Manager	mina@shop.com	Kathmandu	45000.00
109	Hari Bhandari	Receptionist	hari@shop.com	Pokhara	30000.00
110	sita shrestha	Technician	gita@shop.com	Biratnagar	38000.00

Q.N.11. Write a query to remove foreign key from payment.

Query: alter table payment

drop foreign key payment_ibfk_1;

Result:

```
| payment | CREATE TABLE `payment` (
  `PaymentID` int(11) NOT NULL,
  `AppointmentID` int(11) DEFAULT NULL,
  `Mode` varchar(50) DEFAULT NULL,
  `Amount` decimal(10,2) DEFAULT NULL,
  `PaymentDate` date DEFAULT NULL,
  PRIMARY KEY (`PaymentID`),
  KEY `AppointmentID` (`AppointmentID`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci
```

Q.N.12. Write a query to remove primary key of employee table.

Query: alter table employees drop primary key;

Result:

```
| employees | CREATE TABLE `employees` (
  `EmployeeID` int(11) NOT NULL,
  `Name` varchar(100) DEFAULT NULL,
  `Role` varchar(50) DEFAULT NULL,
  `Contact` varchar(100) DEFAULT NULL,
  `Address` varchar(200) DEFAULT NULL,
  `Salary` decimal(10,2) DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci |
```

Q.N.13. Write a SQL statement to make table employee empty.

Query: truncate table Employees;

Result:

```
MariaDB [nationwidedb]> select * from employees;
Empty set (0.001 sec)
```

Q.N.14. Insert id, name, address, email, phone, salary in table teacher and id should be automatic.

Query: create table teacher (

id int auto_increment primary key,

name varchar(100),

address varchar(200),

email varchar(100),

phone varchar(15),

salary decimal(10, 2)

);

insert into teacher (name, address, email, phone, salary) values

('ram thapa', 'kathmandu', 'ram@school.com', '9841234567', 30000.00),

('sita gurung', 'pokhara', 'sita@school.com', '9851234567', 35000.00),

('hari lama', 'biratnagar', 'hari@school.com', '9861234567', 32000.00),

('gita rai', 'dharan', 'gita@school.com', '9871234567', 40000.00),

('mohan shah', 'lalitpur', 'mohan@school.com', '9881234567', 38000.00);

Result:

ID	Name	Address	Email	Phone	Salary
1	Ram Thapa	Kathmandu	ram@school.com	9841234567	30000.00
2	Sita Gurung	Pokhara	sita@school.com	9851234567	35000.00
3	Hari Lama	Biratnagar	hari@school.com	9861234567	32000.00
4	Gita Rai	Dharan	gita@school.com	9871234567	40000.00
5	Mohan Shah	Lalitpur	mohan@school.com	9881234567	38000.00

Q.N.15. Write a SQL statement to display teacher's name whose salary is bigger than average salary.

Query: select name

from teacher where salary > (select avg(salary) from teacher);

Result:

Name
Gita Rai
Mohan Shah

Q.N.16. Display total salary of teacher

table. Query:

```
select sum(salary) as totalsalary
```

```
from teacher;
```

Result:

TotalSalary
175000.00

Q.N.17. Write a SQL statement to count all the number of rows of teacher

table. Query: select count(*) as rowcount

```
from teacher;
```

Result:

RowCount
5

Q.N.18 Write a SQL statement to find the maximum salary of

teacher. Query: select max(salary) as maxsalary

```
from teacher;
```

Result:

MaxSalary
40000.00

Q.N.19. Delete teacher's name from table teacher whose id is

3. Query: delete from teacher where id = 3;

Result:

ID	Name	Address	Email	Phone	Salary
1	Ram Thapa	Kathmandu	ram@school.com	9841234567	30000.00
2	Sita Gurung	Pokhara	sita@school.com	9851234567	35000.00
4	Gita Rai	Dharan	gita@school.com	9871234567	40000.00
5	Mohan Shah	Lalitpur	mohan@school.com	9881234567	38000.00

Q.N.20. Change the name of teacher with “SHYAM” whose id is 5 in teacher

table. Query: UPDATE Teacher SET Name = 'SHYAM' WHERE ID = 5;

Result:

ID	Name	Address	Email	Phone	Salary
1	Ram Thapa	Kathmandu	ram@school.com	9841234567	30000.00
2	Sita Gurung	Pokhara	sita@school.com	9851234567	35000.00
4	Gita Rai	Dharan	gita@school.com	9871234567	40000.00
5	shyam	Lalitpur	mohan@school.com	9881234567	38000.00

Q.N.21. Write a query to create table department where EID is working as a foreign

key. Query: create table department (

did int primary key,

dname varchar(100),

eid int,

foreign key (eid) references employees(employeeid)

);

Result:

```

| department | CREATE TABLE `department` (
  `DID` int(11) NOT NULL,
  `Dname` varchar(100) DEFAULT NULL,
  `EID` int(11) DEFAULT NULL,
  PRIMARY KEY (`DID`),
  KEY `EID` (`EID`),
  CONSTRAINT `department_ibfk_1` FOREIGN KEY (`EID`) REFERENCES `employees` (`EmployeeID`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci

```

Q.N.22. Write a query which will increase the salary of each department by Rs

5000. Query: update employees

set salary = salary + 5000

where department is not null;

Result:

EmployeeID	Name	Role	Contact	Address	Salary	Department
101	Arjun Karki	Technician	arjun@shop.com	Humla District	45000.00	Technical
102	Bimala Rai	Manager	bimala@shop.com	Kathmandu	50000.00	Management
103	Kamal Shah	Technician	kamal@shop.com	Humla	43000.00	Technical
104	Laxmi Adhikari	Receptionist	laxmi@shop.com	Pokhara	35000.00	Customer Service
105	Nabin Poudel	Technician	nabin@shop.com	Biratnagar	40000.00	Sales

Q.N.23. Write a query to increase salary of employees by 5% whose salary is more than 9000.

Query: update employees

set salary = salary * 1.05

where salary > 9000;

Result:

EmployeeID	Name	Role	Contact	Address	Salary	Department
101	Arjun Karki	Technician	arjun@shop.com	Humla District	47250.00	Technical
102	Bimala Rai	Manager	bimala@shop.com	Kathmandu	52500.00	Management
103	Kamal Shah	Technician	kamal@shop.com	Humla	45150.00	Technical
104	Laxmi Adhikari	Receptionist	laxmi@shop.com	Pokhara	36750.00	Customer Service
105	Nabin Poudel	Technician	nabin@shop.com	Biratnagar	42000.00	Sales

Q.N.24. Write a query to change the employee id of employees whose old department id is 2001. The new department id should be 6001.

Query: update employees

set departmentid = 6001

where departmentid = 2001;

Result:

EmployeeID	Name	Role	Contact	Address	Salary	Department	DepartmentID
101	Arjun Karki	Technician	arjun@shop.com	Humla District	47250.00	Technical	6001
102	Bimala Rai	Manager	bimala@shop.com	Kathmandu	52500.00	Management	2002
103	Kamal Shah	Technician	kamal@shop.com	Humla	45150.00	Technical	6001
104	Laxmi Adhikari	Receptionist	laxmi@shop.com	Pokhara	36750.00	Customer Service	2003
105	Nabin Poudel	Technician	nabin@shop.com	Biratnagar	42000.00	Sales	2004